

Emily Qi

emilyhlqi@gmail.com | emilyqi.com | linkedin.com/in/emily-q | github.com/emlyqi

EDUCATION

University of Waterloo

2024 – 2028

Candidate for Bachelor of Computer Science, Honours (3.9 GPA)

- Computing and Financial Management Outstanding Achievement Award

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, HTML, CSS, SQL

Libraries and Frameworks: React, Next.js, Node.js, Express, Django, FastAPI, Flask, PostgreSQL, MongoDB, SQLite, OpenAI, OpenCV, PyTorch, scikit-learn, HuggingFace Transformers, ONNX, pandas, NumPy, Selenium

Tools: Git, Docker, Firebase, Google Cloud, Cloudflare, Vercel, Railway, Weights & Biases, Postman, Figma

EXPERIENCE

Machine Learning Engineering Intern

Jan 2026 – present

HubHead

- Developed a visual grounding validation system for 250+ document types: generates field descriptions from bounding box annotations via Gemini 2.5 Pro, predicts entity positions from text alone, and iteratively refines via IoU against ground-truth for per-edge spatial feedback
- Built a multimodal template recommender using Gemini 768-dim embeddings with statistical ranking (z-score normalization, clarity-weighted aggregation, consensus scoring), adaptive subsampling, and <1.0s cached inference
- Scaled ML extraction pipelines to 70K+ pages across Document AI, Gemini, and Textract on Cloud Run, offloading to a dedicated worker service with streaming ingestion, semaphore-gated concurrency, heartbeat fault recovery, and real-time team collaboration via Flask-SocketIO

Machine Learning Researcher

Sep 2025 – Dec 2025

University of Waterloo

- Built an end-to-end Soft Actor-Critic (SAC) reinforcement learning pipeline for autonomous racing in Trackmania with Stable Baselines 3, OpenAI Gym, and TMRL's OpenPlanet telemetry interface
- Designed a multi-component reward function combining track progress, speed incentives, and action smoothness penalties with tunable weights, and tracked training runs with Weights & Biases

Software Engineering Intern

May 2025 – Aug 2025

Royal Canadian Mounted Police

- Drove end-to-end development of a full-stack app (React, Next.js, FastAPI, SQLite) using RESTful APIs and CRUD operations for asynchronous execution of key operational workflows, reducing manual overhead by 70%
- Led client conversations to translate business needs into product requirements, accelerating delivery by 2 months

PROJECTS

Depth Estimation Benchmark | *Python, PyTorch, OpenCV, HuggingFace Transformers, ONNX, NumPy, W&B*

- Benchmarked stereo and neural depth estimation on 200 KITTI scenes; fine-tuned DPT-Large with Huber loss, improving AbsRel by 19% and surpassing the stronger pretrained DepthAnything V2 on MAE/RMSE
- Exported fine-tuned DPT-Large to ONNX by patching ViT source to resolve symbolically dynamic tensor shapes blocking export; applied INT8 quantization achieving 4× model size reduction (1368MB → 352MB)

Karyon | *Django, React, OpenAI, sentence-transformers, OpenCV, Vercel, Railway, Docker, PostgreSQL*

- Developed full-stack **video Q&A platform** enabling natural language queries with timestamp-linked citations and multimodal processing (audio, visual, hybrid), supporting uploads and YouTube imports
- Engineered a RAG pipeline combining Whisper transcription, GPT-4o vision keyframe analysis, and semantic chunking with sentence-transformer embeddings; used OpenCV change detection to reduce vision API calls by 80%
- Containerized and deployed to production with JWT auth, Fernet-encrypted API keys, and per-user data isolation

Atari Breakout with DQN | *Python, PyTorch, Deep Reinforcement Learning, OpenCV, Weights & Biases*

- Implemented and combined three deep RL papers (DQN, Double DQN, Prioritized Experience Replay) from scratch in PyTorch; trained agent for 10M frames on Atari Breakout, achieving 7× reward improvement and 227× Q-value growth over random baseline
- Built memory-efficient prioritized replay buffer with uint8 frame compression (4× storage reduction), TD-error prioritization, and importance sampling bias correction for stable training on constrained GPU hardware